

Practice Sheet 1
CSE/ECE 511 Computer Architecture

Q1.

Which of the following snippets of code has better temporal locality. Explain.

a.

```
for(i = 0; i < 10; i++)  
    for(j = 0; j < 10; j++)  
        a[j] = b[j];
```

b.

```
for(j = 0; j < 10; j++)  
    for(i = 0; i < 10; i++)  
        a[j] = b[j];
```

Solution: In part A, the array is indexed by the fastest moving variable. The access pattern looks like

A[0], A[1], A[2], ..., A[9], A[0], A[1], ..., A[9], ...

In part B, the array is indexed by the slowest moving variable. The access pattern looks like A[0], A[0], ... A[0], A[1], A[1], ..., A[1], ...

Therefore pattern B has better temporal locality as the same address is accessed again and again.

Q2. Consider a processor with 5 pipeline stages. The delay of each pipeline stage is 4 ns, 5ns, 10 ns, 10 ns, and 7ns respectively. The registers residing between the pipeline stages have a delay of 2 ns. Determine the approximate speedup of the pipelined implementation as compared to its non-pipelined implementation?

Solution:

Time taken to execute N instructions in non-pipelined implementation = $(4 + 5 + 10 + 10 + 7)N = 36N$

Time taken to execute N instructions in pipelined implementation = $(\max\{4,5,10,10,7\} + 2)N = 12N$.

Speedup = $36N / 12N = 3$

Q3

Consider a 32-bit machine with a set-associative cache. The total cache size is 16KB and it is a 4-way set-associative cache. The length of the memory location is 1 byte and the block size is 16 Bytes.

- a) Calculate the total number of sets present in the cache.
- b) Draw the 32-bit memory address showing the tag, index, and offset bits.
- c) Determine the set number where the content of memory address 10FFCFF(HEX) is mapped. Give an explanation or steps.

Solution

- a) $n=32$
 $M=16KB=2^{14}B=2^{14}B$
 $block=B=16B=2^4B$
 $E=4=2^2$
 $M=S \times E \times B$
 $S=M \div (E \times B)=2^{14} \div (2^2 \times 2^4)=2^{14-6}=2^8=256$ sets
- b) tag = 20
Index= 8
Offset = 4
- c) 207.

Q4

- a) Two processors A and B are working on the same ISA. The frequency of processor A is higher than the frequency of processor B. For a given program, does it imply that processor A always executes more instruction per second than processor B?
- b) Two processors A and B are working on the same ISA. For a given program, if A executes x Instructions per second, and B executes y instructions per second. If $x > y$, does this imply processor A is faster than processor B?

Solution

- a) No, Processor B might have a much higher IPC (instructions per cycle).
For example, a processor with a lower frequency might be able to execute multiple instructions per cycle while a processor with a higher frequency might only execute one instruction per cycle.
- b) Yes, as the number of instructions is the same.

Q5

Consider a fully bypassed 5-stage RISC-V processor with the following pipeline stages: F, D, X, M, W. Whenever a branch instruction is encountered, it is resolved in the E Stage of the pipeline. The F Stage always speculates that the next PC is PC+4. Here, concentrate on only conditional branches and ignore any unconditional jumps. Consider the following set of instructions given below. Note that the value of register R0 is 0.

0x1000: ADDI R4, R0, 9

0x1004: LD R5, 0(R0)

0x1008: ADD R5, R5, R4

0x100C: BEQ R4, R5, 0x1000

0x1010: LD R7, 4(R6)

0x1014: ADD R5, R7, R5

0x1018: SUB R1, R2, R3

Draw the pipeline timing diagram (resource usage diagram will not be considered) of the above instructions. You can draw the pipeline till seven instructions. Calculate how many clock cycles will be required to complete till seven instructions.

Solution:

PC	Instruction	1	2	3	4	5	6	7	8	9	10	11	12
0x1000	ADDI R4, R0, 9	F	D	X	M	W							
0x1004	LD R5, 0(R0)		F	D	X	M	W						
0x1008	ADD R5, R5, R4			F	D	D	X	M	W				
0x100C	BEQ R4, R5, 0x1000				F	F	D	X	M	W			
0x1010	LD R7, 4(R6)						F	D	-	-	-		
0x1014	ADD R5, R7, R5							F	-	-	-	-	
0x1018	SUB R1, R2, R3								F	D	X	M	W

Q6

For a memory chip with 256 addresses, where each address represents a space of 8 bits, what are the address bus and data bus size required to access the whole memory? Also, how many 4-bit values can we store in the given memory chip?

Solution

Let size of address bus = S

$$\Rightarrow 2^S = 256 \text{ address}$$

$$\Rightarrow 2^S = 2^8$$

$$\Rightarrow S = 8$$

Let the size of data bus = D

D requires to send 1 byte of data which will be 8 bits

$$D = 8 \text{ bits}$$

We can break each memory location into two 4-bit chunks. Therefore, we can store 512 4-bit values.

Q7

Classify the following attributes of a machine as either a property of its microarchitecture or ISA. Clearly state your reason for each property.

Attribute	ISA?	Microarchitecture?
The latency of each pipeline stage in the machine.		
Addressing modes available for arithmetic operations		
A 5-bit immediate can be specified in an ADD instruction		
The machine has multi-level cache hierarchy		
The width of general-purpose registers		
ALU bypasses its result to a 2-to-1 mux which feeds its output to one of the inputs of ALU		
The machine converts CISC instructions to RISC instructions before starting executing the instruction		

Solution

Attribute	ISA?	Microarchitecture?
The latency of each pipeline stage in the machine.		✓
Addressing modes available for arithmetic operations	✓	
A 5-bit immediate can be specified in an ADD instruction	✓	
The machine has multi-level cache hierarchy		✓
The width of general-purpose registers	✓	
ALU bypasses its result to a 2-to-1 mux which feeds its output to one of the inputs of ALU		✓
The machine converts CISC instructions to RISC instructions before starting executing the instruction		✓

Q8

Suppose you have a 32-bit processor that has a byte-addressable memory. This processor has a 512 byte 2-way set-associative cache, with 16-byte cache lines. The cache uses LRU (least recently used) replacement policy. Split the 32-bit address into "tag", "index", and "cache-line offset" pieces. Which address bits comprise each piece?

- What is the total number of cache lines?
- How many sets does this cache have?
- How many cache-line are in each set?
- How many bits are required for a tag, index, and cache-line offset?
- Mark which segments of the address denote tag, index, and cache-line offset.

- f) The memory read access trace is given below. Assume that initially, the cache is empty. Classify each reference as a hit or miss. When it is a miss, classify the misses as compulsory, conflict, or capacity. First has been solved for your reference. Assume that there are no write accesses made to the cache.

Address	Hit/Miss	Miss type
0x0300	Miss	Compulsory
0x0310		
0x0401		
0x0502		
0x0501		
0x0301		
0x040C		
0x030C		

- g) Calculate the miss rate and hit rate.

- h) For the given memory read access trace, write the address range stored in the cache line of set-0 after the given memory access. Also, mark the cache line which will be evicted when the next conflict miss occurs i.e. LRU cache line. One is filled for you as an example.

Memory access	Set 0, Way 1(cache line 1)	Set 0, Way 2(cache line 2)
0x0300(set 0)	0x0300-0x030F	random_value(LRU)
0x0310(set 1)		
0x0401(set 0)		
0x0502(set 0)		
0x0501(set 0)		
0x0301(set 0)		
0x040C(set 0)		
0x030C(set 0)		

Solution

- $512/16 = 32$
- Number of cache lines / number of ways = $32 / 2 = 16$
- 2 (As it is a 2-way set-associative cache)
- Index - 4, cache line offset - 4, tag = $32 - 8 = 24$
- The address will be broken down in the following manner:

Tag (24 bits)	Index (4 bits)	Offset (4 bits)
32	8	4
		0

f) Memory Access Tracking:

Address	Hit/Miss?	Miss type
0x0300	Miss	Compulsory (set 0)
0x0310	Miss	Compulsory (set 1)
0x0401	Miss	Compulsory (set 0)
0x0502	Miss	Compulsory (set 0) (0x0300-0x030F evicted)
0x0501	Hit	(set 0) (data already present when 0x0502 was accessed)
0x0301	Miss	Conflict (set 0) (0x0400-0x040F evicted)
0x040C	Miss	Conflict (set 0) (0x0500-0x050F evicted)
0x030C	Hit	(data already in cache when 0x0301 was accessed)

g) hit rate = $2/7$ miss rate = $1 - 2/7 = 5/7$

h) LRU Tracking:

We are showing the values *after* the cache is accessed by the CPU.

Memory access	Set 0, Way 1(cache line 1)	Set 0, Way 2(cache line 2)
0x0300(set 0)	0x0300-0x030F	random_value(LRU)
0x0310(set 1)	0x0300-0x030F	random_value(LRU)
0x0401(set 0)	0x0300-0x030F(LRU)	0x0400-0x040F
0x0502(set 0)	0x0500-0x050F	0x0400-0x040F(LRU)
0x0501(set 0)	0x0500-0x050F	0x0400-0x040F(LRU)
0x0301(set 0)	0x0500-0x050F(LRU)	0x0300-0x030F
0x040C(set 0)	0x0400-0x040F	0x0300-0x030F(LRU)
0x030C(set 0)	0x0400-0x040F(LRU)	0x0300-0x030F

Q9

Consider the following characteristics of Machine A and Machine B with the same instruction set. The clock rate of A and B are 80 MHz and 100 MHz respectively with the following set of instructions given in the table below.

Instruction type	CPI for Machine A	CPI for Machine B	% of each instruction type
ALU	1	2	60 %
Load/Store	2	3	30 %
Branches	4	4	10 %

- a) Calculate the average CPI of Machine A and B.
- b) Calculate the average MIPS (Million Instructions Per Second) rate for Machine A and B.

Solution

- a) Machine A = $(60/100)*1 + (30/100)*2 + (10/100)*4 = 1.6$
Machine B = $(60/100)*2 + (30/100)*3 + (10/100)*4 = 2.5$
- b) For machine A
 $MIPS = \text{Clock rate}/CPI * 10^6 = 80 * 10^6 / 1.6 * 10^6 = 50$
For machine B
 $MIPS = \text{Clock rate}/CPI * 10^6 = 100 * 10^6 / 2.5 * 10^6 = 40$